

Symmetric-determinant apolarity in odd characteristic

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Abstract

Let D_N be the determinant of the generic symmetric $N \times N$ matrix, and let differential operators act by ordinary differentiation over a field of odd characteristic p . We determine the apolar ideal of D_N in every size. In addition to the classical quadratic relations, it is generated by one $(p-1) \times (p-1)$ determinant of differential variables for each set of $2p-2$ indices. These additional generators are minimal modulo the quadratic ideal. An exterior-algebra realization identifies the remaining apolar relations with radicals of invariant pairings for SL_2 . Semisimplification gives a Hilbert series in terms of Dyck paths of height at most $p-2$, and the Steinberg symmetrizer gives the explicit generators of the entire radical. We also compute the ranks of powers of a general linear form. The weak Lefschetz property holds exactly when $N \leq 2p-2$, and the strong Lefschetz property holds exactly when $N < p$.

1 Introduction and main results

Let K be a field and let $X = (x_{ij})_{1 \leq i, j \leq N}$ be the generic symmetric matrix, where $x_{ji} = x_{ij}$ and the variables x_{ij} with $i \leq j$ are algebraically independent. Set

$$R = K[x_{ij} : 1 \leq i \leq j \leq N], \quad S = K[d_{ij} : 1 \leq i \leq j \leq N], \quad D_N = \det X.$$

Throughout, d_{ij} acts on R as the ordinary partial derivative $\partial/\partial x_{ij}$, and d_{ji} means d_{ij} . The apolar algebra is

$$A_N = S / \mathrm{Ann}(D_N), \quad \mathrm{Ann}(D_N) = \{f \in S : f(\partial)D_N = 0\}.$$

The grading on S assigns degree one to every d_{ij} . We assume $N \geq 1$.

In characteristic zero, Shafiei determined the apolar ideal of the symmetric determinant and proved that it is generated by explicit quadrics [S, Theorem 3.10]. Ordinary differentiation in positive characteristic introduces a further issue: the top multiplication pairing on the quadratic quotient can become degenerate. The first degeneracy in characteristic $p > 2$ occurs at $N = 2p-2$. We describe this degeneracy in all degrees and show that its first occurrence generates all subsequent relations.

Write $Q_N \subset S$ for the ideal generated by

$$\begin{aligned} & d_{ii}^2, \quad d_{ii}d_{ij} \quad (i \neq j), \\ & d_{ij}^2 + 2d_{ii}d_{jj} \quad (i \neq j), \\ & d_{ij}d_{ik} + d_{ii}d_{jk} \quad (i, j, k \text{ distinct}), \\ & d_{ij}d_{kl} + d_{ik}d_{jl} + d_{il}d_{jk} \quad (i, j, k, l \text{ distinct}). \end{aligned} \tag{1.1}$$

Suppose $\text{char } K = p > 2$ and put $m = p - 1$. For a subset $U = \{u_1 < \cdots < u_{2m}\} \subseteq [N]$, define

$$F_U = \det(d_{u_i, u_{m+j}})_{1 \leq i, j \leq m}. \quad (1.2)$$

Thus the row and column index sets are disjoint and together form U .

Theorem 1.1 (Apolar ideal). *Over a field of odd characteristic p ,*

$$\text{Ann}(D_N) = Q_N + (F_U : U \subseteq [N], |U| = 2p - 2). \quad (1.3)$$

The classes of the displayed F_U form a basis of $(\text{Ann}(D_N)/Q_N)_{p-1}$ and minimally generate $\text{Ann}(D_N)/Q_N$ as an S -module. In particular, there are exactly $\binom{N}{2p-2}$ additional generators modulo Q_N . If $p \geq 5$, the apolar ideal is generated by quadrics if and only if $N < 2p - 2$; its only possible minimal generator degrees are 2 and $p - 1$. In characteristic 3, the entire apolar ideal is generated by quadrics. In characteristic zero, $\text{Ann}(D_N) = Q_N$.

A Dyck path of length $2s$ is a walk from 0 to 0 with steps $+1$ and -1 which never goes below 0. Let $b_s(p)$ denote the number of these paths whose height is at most $p - 2$, and put $b_0(p) = 1$. We use $\binom{a}{b} = 0$ when $b < 0$ or $b > a$, for $a \geq 0$.

Theorem 1.2 (Hilbert series). *Over a field of odd characteristic p , the Hilbert series of A_N is*

$$H_{A_N}(t) = \sum_{s=0}^{\lfloor N/2 \rfloor} \binom{N}{2s} b_s(p) t^s (1+t)^{N-2s}. \quad (1.4)$$

If P is the adjacency matrix of the path with vertices $0, 1, \dots, p-2$, then

$$\dim_K A_N = ((2I + P)^N)_{0,0} = \sum_{j \in \mathbb{Z}} \left[\binom{2N}{N+2pj} - \binom{2N}{N+2pj+2} \right]. \quad (1.5)$$

The sum is finite under the binomial convention above. In characteristic zero, the same Hilbert-series formula holds with $b_s(0) = \text{Cat}_s = \frac{1}{s+1} \binom{2s}{s}$; in that case

$$\dim_K (A_N)_k = \frac{1}{N+1} \binom{N+1}{k} \binom{N+1}{k+1}, \quad \dim_K A_N = \text{Cat}_{N+1}. \quad (1.6)$$

For a standard graded Artinian algebra A , a linear form $L \in A_1$ is weak Lefschetz if multiplication $A_k \xrightarrow{L} A_{k+1}$ has maximal rank for every k . It is strong Lefschetz if $A_k \xrightarrow{L^q} A_{k+q}$ has maximal rank for every $k, q \geq 0$. The corresponding properties mean that such a form exists over the ground field.

Theorem 1.3 (Lefschetz properties). *Over any field of odd characteristic p , the algebra A_N has the weak Lefschetz property exactly when $N \leq 2p - 2$, and it has the strong Lefschetz property exactly when $N < p$. Whenever either property holds, it is realized by*

$$\ell = d_{11} + \cdots + d_{NN}.$$

At $N = 2p - 1$, the map $\ell : (A_N)_{p-1} \rightarrow (A_N)_p$ has a one-dimensional kernel. In characteristic zero, ℓ is strong Lefschetz for every N .

The proof separates a characteristic-independent quadratic algebra from the characteristic-dependent radical of its top pairing. The former is a subalgebra of an exterior algebra. Its multigraded pieces are spaces of invariant tensors for the standard representation of SL_2 . The rank of the invariant pairing is determined by the semisimplification of tilting modules. To obtain ideal generators, we use the Steinberg symmetrizer and then translate its tensor-ideal property into multiplication in S/Q_N . Finally, multiplication by ℓ decomposes into subset-inclusion maps, whose ranks are given by Wilson's formula in the form proved by Frankl [F, Theorem 4.1].

The ordinary differential convention is essential. The characteristic assumption stated in [S] includes all $p > 2$, whereas the quadratic presentation requires the additional relations in (1.3) in small characteristic. We use its characteristic-zero presentation as the point of comparison. Section 5 gives the first explicit discrepancy. No divided-power action is used anywhere in this paper.

2 The exterior model and its quadratic presentation

Assume for this section that 2 is invertible in K . Let

$$\Lambda = \bigwedge \left(\bigoplus_{i=1}^N (K a_i \oplus K b_i) \right), \quad u_i = a_i b_i, \quad z_{ij} = a_i b_j + a_j b_i \quad (i \neq j).$$

All products in Λ are exterior products; we omit wedge symbols. The elements u_i and z_{ij} have even exterior degree and commute. There is therefore an algebra homomorphism

$$\phi : S \longrightarrow \Lambda, \quad d_{ii} \longmapsto u_i, \quad d_{ij} \longmapsto z_{ij} \quad (i < j). \quad (2.1)$$

Give a_i, b_i vertex degree $e_i \in \mathbb{Z}^N$. On S this means $\deg_v d_{ii} = 2e_i$ and $\deg_v d_{ij} = e_i + e_j$. All the relations in (1.1) are homogeneous for this grading.

For a finite ordered set E of even size $2s$, let $W(E)$ be the span of the products

$$z_M = \prod_{\{i,j\} \in M} z_{ij},$$

where M runs through the perfect matchings of E . The space $W(E)$ lies in the exterior component of vertex degree one at each vertex of E and zero elsewhere. Write $u_T = \prod_{i \in T} u_i$.

Proposition 2.1 (Quadratic normal form). *The kernel of ϕ is Q_N . If $C_N = S/Q_N$, then its vertex-multigraded pieces are*

$$(C_N)_\alpha = \begin{cases} u_T W(E), & \alpha = \mathbf{1}_E + 2\mathbf{1}_T, \quad E \cap T = \emptyset, \quad |E| = 2s, \\ 0, & \text{otherwise.} \end{cases} \quad (2.2)$$

The products associated with noncrossing matchings of E form a basis of $W(E)$. Consequently $\dim_K W(E) = \mathrm{Cat}_s$, independently of the odd characteristic.

Proof. Exterior multiplication gives the relations in (1.1). For example,

$$z_{ij}^2 = -2u_i u_j, \quad z_{ij} z_{jk} = -u_j z_{ik}, \quad z_{ij} z_{kl} + z_{ik} z_{jl} + z_{il} z_{jk} = 0.$$

This proves $Q_N \subseteq \ker \phi$.

Represent a monomial in S by a multigraph on $[N]$, with a loop at i for each factor d_{ii} and an edge ij for each factor d_{ij} . A loop contributes two to the vertex degree. A monomial with vertex degree at least three at some vertex vanishes modulo Q_N . Indeed, a loop together with any further incident factor is zero. A repeated edge can be replaced by $-2d_{ii}d_{jj}$, reducing to that case. For three distinct edges incident at i , replacing the first two by $-d_{ii}d_{jk}$ again gives zero.

If every vertex degree is at most two, each nontrivial component is a path, a cycle, or a loop. Repeated use of $d_{ij}d_{jk} = -d_{jj}d_{ik}$ reduces a path to its endpoint edge times the diagonal variables at its internal vertices, with a sign. A cycle reduces to a signed factor 2 times the product of the diagonal variables at its vertices. It follows that the component of vertex degree $\mathbf{1}_E + 2\mathbf{1}_T$ is spanned by $(\prod_{i \in T} d_{ii})(\prod_{\{i,j\} \in M} d_{ij})$, with M a matching on E .

The four-vertex relation straightens these matching products into noncrossing ones. This is the usual bracket straightening: place the ordered vertices on a line and resolve each crossing by the other two pairings of its four ends. Drawing the resolved arcs before straightening them removes a crossing, so iteration terminates. Equivalently, this is the multilinear Plücker straightening for two-dimensional brackets.

For independence, order the exterior generators by vertex. In the component with one generator at each vertex, identify a monomial with its word in a, b . With $a > b$ in lexicographic order, the leading word of z_M has a at each left endpoint and b at each right endpoint, with coefficient 1 or -1 . Noncrossing matchings have distinct leading words: these are exactly the Dyck words, and the matching is recovered by the usual stack pairing. Their images are therefore linearly independent in every characteristic. Multiplication by u_T on disjoint vertices preserves independence. The spanning set just obtained in S/Q_N consequently maps to a basis of the image, proving the claim. $\square$

Let $V = Ka \oplus Kb$ with the symplectic form $\varepsilon(a, b) = 1$. Its symplectic group is SL_2 . Ordered exterior multiplication identifies the component with one letter at each vertex of E with $V^{\otimes 2s}$. Under this identification z_{ij} is the alternating invariant tensor on positions i, j , with the signs supplied by the order of the exterior factors. The characteristic-free first fundamental theorem for symplectic invariants gives

$$W(E) = (V^{\otimes 2s})^{\mathrm{SL}_2}. \quad (2.3)$$

One may apply the multilinear part of [H, Theorem 17]; in dimension two its generators are the brackets used above. We shall also use its consequence that equivariant maps between tensor powers of V are spanned by diagrams made from symplectic evaluations, coevaluations, and permutations of factors. This follows by identifying V with V^* and bending input factors into output factors.

3 The apolar pairing and its modular rank

Let $\tau : \Lambda \rightarrow K$ extract the coefficient of $u_{[N]} = u_1 \cdots u_N$. Consider the finite expression

$$\mathcal{E}(X) = \prod_{i=1}^N (1 + x_{ii}u_i) \prod_{1 \leq i < j \leq N} \left(1 + x_{ij}z_{ij} + \frac{x_{ij}^2 z_{ij}^2}{2} \right). \quad (3.1)$$

It is an element of $R \otimes_K \Lambda$. The quadratic denominators are units under our characteristic assumption.

Lemma 3.1. *The identities*

$$\tau(\mathcal{E}(X)) = D_N, \quad f(\partial)D_N = \tau(\phi(f)\mathcal{E}(X)) \quad (f \in S) \quad (3.2)$$

hold over every field of characteristic different from two. Consequently

$$\text{Ann}(D_N)/Q_N = \{c \in C_N : \tau(cc') = 0 \text{ for all } c' \in C_N\}. \quad (3.3)$$

Proof. The determinant identity can be checked over $\mathbb{Q}$ by writing

$$\mathcal{E}(X) = \exp \left(\sum_i x_{ii} a_i b_i + \sum_{i < j} x_{ij} (a_i b_j + a_j b_i) \right).$$

The coefficient of $u_{[N]}$ is the usual alternating permutation expansion of $\det X$. The finite product (3.1) and this coefficient identity are defined over $\mathbb{Z}[1/2]$, so the identity holds after specialization to every field under consideration. In particular, no exponential series with noninvertible factorials is being used in positive characteristic.

Direct differentiation of each factor in (3.1) gives $\partial_{ij}\mathcal{E} = \phi(d_{ij})\mathcal{E}$, since $u_i^2 = 0$ and $z_{ij}^3 = 0$. Iteration proves the second identity in (3.2).

For the last assertion, let

$$B = \{\beta : \beta_{ii} \in \{0, 1\}, \beta_{ij} \in \{0, 1, 2\} \text{ for } i < j\}, \quad \beta! = \prod_{i \leq j} \beta_{ij}!.$$

Expanding the finite product gives

$$\mathcal{E}(X) = \sum_{\beta \in B} \frac{x^\beta}{\beta!} \phi(d^\beta), \quad f(\partial)D_N = \sum_{\beta \in B} \frac{x^\beta}{\beta!} \tau(\phi(f)\phi(d^\beta)).$$

Every $\beta!$ in these sums is a power of 2 and hence is a unit. Since the distinct monomials x^β are linearly independent in the polynomial ring R , the second sum vanishes if and only if $\tau(\phi(f)\phi(d^\beta)) = 0$ for every $\beta \in B$. This is a coefficientwise statement about formal polynomials, valid also over a finite field. The elements $\phi(d^\beta)$ with $\beta \in B$ span C_N : every operator monomial outside these exponent bounds has zero image, by $u_i^2 = 0$ and $z_{ij}^3 = 0$. Thus $f(\partial)D_N = 0$ if and only if $\phi(f)$ pairs to zero with every element of C_N . Together with $\ker \phi = Q_N$, this proves (3.3). $\square$

The radical in (3.3) is an ideal, by associativity of multiplication. Only complementary vertex degrees can pair under τ . If $\alpha = \mathbf{1}_E + 2\mathbf{1}_T$, its complement in $2\mathbf{1}_{[N]}$ is $\mathbf{1}_E + 2\mathbf{1}_{[N] \setminus (E \cup T)}$. To compute this pairing, order the letters at the vertices of E and move the two letters at each vertex next to one another. This reordering contributes a fixed sign, independent of the letters; the coefficient of $a_i b_i$ at each vertex is the corresponding value of ε . The induced form on $W(E)$ is therefore, up to an overall sign, the restriction of $\varepsilon^{\otimes 2s}$. Let $\mathcal{R}(E)$ denote its radical and put

$$U(E) = W(E)/\mathcal{R}(E).$$

It follows that

$$(A_N)_{\mathbf{1}_E + 2\mathbf{1}_T} \simeq u_T U(E). \quad (3.4)$$

In particular, the quotient has a nondegenerate top pairing and one-dimensional top degree N , generated by $u_{[N]}$. This also proves directly that A_N is Artinian Gorenstein in the ordinary differential convention used here.

We now compute $\dim U(E)$. The representation-theoretic argument may be made over an algebraic closure: the ranks of the matrices defining all the spaces and pairings above are unchanged by extending scalars.

Proposition 3.2. *If $|E| = 2s$ and $\text{char } K = p > 2$, then*

$$\dim_K U(E) = b_s(p).$$

In characteristic zero, the invariant pairing is nondegenerate and $\dim_K U(E) = \text{Cat}_s$.

Proof. Suppose first that K is algebraically closed of characteristic p . Let T_r be the indecomposable tilting SL_2 -module of highest weight r . Recall that a morphism $f : M \rightarrow M'$ is negligible if $\text{tr}(gf) = 0$ for every $g : M' \rightarrow M$. The negligible tilting modules are exactly the direct sums of T_r with $r \geq p - 1$, and a morphism between tilting modules is negligible precisely when it factors through such a module [BFRSW, Lemma 2.6 and Corollary 2.7]. The resulting quotient category is semisimple, with simple objects the images of $T_0, \dots, T_{p-2}$; these modules are $\text{Sym}^r V$ in the indicated range. For the general semisimplification framework, see also [BEE0, Sections 1–2].

Set $M = V^{\otimes 2s}$, a tilting module. The pairing between $\text{Hom}(K, M)$ and $\text{Hom}(M, K)$ is composition. Via the symplectic self-duality of V , it is the invariant pairing on $W(E)$. Its radical is therefore exactly the space of negligible morphisms from K to M . The dimension of $U(E)$ is the multiplicity of the unit object in the image of M in the quotient category.

For $1 \leq r \leq p - 2$, the Clebsch–Gordan splitting gives

$$V \otimes \text{Sym}^r V \simeq \text{Sym}^{r+1} V \oplus \text{Sym}^{r-1} V. \quad (3.5)$$

The usual multiplication and polarization maps split this decomposition after division by $r + 1$, which is invertible throughout this range. At $r = p - 2$, the first summand is T_{p-1} and becomes zero in the quotient. At $r = 0$, tensoring by V gives only T_1 . Thus tensoring by V in the quotient has adjacency matrix P on the vertices $0, \dots, p - 2$. The multiplicity in question is $(P^{2s})_{0,0}$, which counts the required bounded Dyck paths.

In characteristic zero, complete reducibility makes the invariant pairing nondegenerate. The unrestricted Clebsch–Gordan rule counts all Dyck paths and gives Cat_s . The conclusions descend from the algebraic closure by invariance of matrix rank under field extension. $\square$

4 Explicit generators of the radical

Throughout this section $\text{char } K = p > 2$ and $m = p - 1$. Fix the ordered tensor basis of $V^{\otimes 2m}$ and define

$$\Omega_m = \sum_{\substack{w \in \{a,b\}^{2m} \\ \#a(w) = \#b(w) = m}} w. \quad (4.1)$$

The tensor is symmetric under every permutation of its factors.

Lemma 4.1 (Steinberg tensor). *The tensor Ω_m is a nonzero invariant tensor. It is a nonzero scalar multiple of the tensor obtained by bending the input factors of the symmetrizer*

$$e_m = \frac{1}{m!} \sum_{\sigma \in \mathfrak{S}_m} \sigma \in \text{End}_{\text{SL}_2}(V^{\otimes m}).$$

It is negligible as a morphism $K \rightarrow V^{\otimes 2m}$. Contraction of any two of its factors by ε is zero.

Proof. Since $m!$ is invertible, e_m is the projector onto $\text{Sym}^m V = T_m$, the Steinberg module. Bend its input factors using $V \simeq V^*$. Up to a fixed overall sign, the coefficient of a balanced word having k letters a in the first half is

$$\frac{(-1)^{m-k}}{\binom{m}{k}}.$$

Indeed, among the $m!$ permutations in the symmetrizer, exactly $k!(m-k)!$ produce a given arrangement with that distribution of letters; the symplectic identification supplies the displayed sign. The congruence $\binom{p-1}{k} = (-1)^k$ in K shows that all balanced words have the same nonzero coefficient. Unbalanced words have coefficient zero. The bent tensor is therefore a nonzero multiple of Ω_m .

The identity of T_m is negligible, so e_m and its bent tensor are negligible as well. Finally, interchanging any selected pair of factors fixes Ω_m and reverses the sign of contraction by the alternating form ε . The contraction is its own negative and hence is zero. $\square$

For $J \subseteq E$ of size $2m$, write Ω_J for the copy of Ω_m on the ordered positions of J . Products with matching tensors on $E \setminus J$ mean tensor products inserted in their specified positions; an exterior reordering can change an overall sign only.

Proposition 4.2 (The entire invariant radical). *For every ordered set E of size $2s$,*

$$\mathcal{R}(E) = \text{span}_K \{ \Omega_J z_M : J \subseteq E, |J| = 2m, M \text{ a matching of } E \setminus J \}. \quad (4.2)$$

In particular, $\mathcal{R}(E) = 0$ if $s < m$, and it is one-dimensional if $s = m$.

Proof. We may again extend scalars to an algebraic closure. By [BFRSW, Lemma 2.6 and Corollary 2.7], a negligible morphism between tilting modules factors through a finite direct sum of modules T_r with $r \geq m$. Each such T_r is a direct summand of

$$T_m \otimes V^{\otimes(r-m)}. \quad (4.3)$$

Indeed, this is a tilting module of highest weight r , and its weight r space is one-dimensional. Its decomposition into indecomposable tilting modules must consequently contain T_r once. Inserting the inclusion and projection of this summand shows that every negligible morphism between tensor powers is a sum of morphisms factoring through modules of the form (4.3).

Apply this observation to an invariant in $\mathcal{R}(E)$, regarded as a negligible morphism $K \rightarrow V^{\otimes 2s}$. Choose equivariant maps

$$i : T_m \longrightarrow V^{\otimes m}, \quad \pi : V^{\otimes m} \longrightarrow T_m, \quad \pi i = 1_{T_m}, \quad i\pi = e_m.$$

For a summand $f = ba$ factoring through $T_m \otimes V^{\otimes q}$, put $h = (i \otimes 1)a$ and $g = b(\pi \otimes 1)$. Then $f = g(e_m \otimes 1)h$, giving the factorization

$$K \xrightarrow{h} V^{\otimes(m+q)} \xrightarrow{e_m \otimes 1} V^{\otimes(m+q)} \xrightarrow{g} V^{\otimes 2s}, \quad (4.4)$$

where h, g are equivariant. Identifying V with V^* , the multilinear part of [H, Theorem 17] expresses both maps as linear combinations of pairing diagrams, as in (2.3). By Lemma 4.1, bending the e_m box replaces it by a nonzero scalar multiple of Ω_m . Absorb this scalar into the coefficient of the diagram. All remaining operations are symplectic evaluations, coevaluations, and permutations of tensor factors.

For completeness, the resulting contractions can be reduced as follows. Away from the distinguished tensor Ω_m and the output positions, every strand has two ends and no branching. Successive evaluations and coevaluations along a strand cancel by the duality identities; using the alternating self-duality of V can introduce only an overall sign. Thus each component is a closed loop or a path whose endpoints are legs of Ω_m or output positions. A path joining two legs of Ω_m contracts those factors by ε , up to sign, and gives zero by Lemma 4.1. In every surviving term all $2m$ legs therefore reach distinct output positions, forming a subset J . The other output positions are paired among themselves, giving a matching M of $E \setminus J$, while closed loops contribute scalars. The term is consequently a scalar multiple of $\Omega_J z_M$. This proves the inclusion from left to right.

Conversely, every tensor on the right is obtained from the bent Steinberg projector by tensoring with invariant pairings and permuting factors. Negligible morphisms form a tensor ideal, so this tensor is negligible and belongs to $\mathcal{R}(E)$. For $s < m$ no such subset J exists; for $s = m$ the right side is the line spanned by the nonzero Ω_m . Equality of the spaces descends to K . $\square$

The preceding proposition describes an invariant tensor space. To obtain generators in the commutative polynomial ring, we identify its distinguished tensor with an actual polynomial in the differential variables.

Lemma 4.3. *For every $U \subseteq [N]$ of size $2m$, the exterior image $\phi(F_U)$ is a nonzero scalar multiple of Ω_U .*

Proof. Relabel U increasingly as $1, \dots, 2m$, with halves L and R . Expanding the determinant and ordering all exterior letters by vertex, the coefficient of a balanced word with k letters a in L is

$$(-1)^{m(m-1)/2+m-k} k!(m-k)!. \quad (4.5)$$

To see this, such a word can occur only when the k positions carrying a in L are matched with the k positions carrying b in R . There are $k!(m-k)!$ compatible bijections. The sign of each bijection in the determinant cancels the corresponding reordering sign on the R positions. Moving the L letters ahead of the R letters contributes $(-1)^{m(m-1)/2}$, and the reversed bracket choices contribute $(-1)^{m-k}$. This gives (4.5).

In characteristic p , the identity

$$k!(m-k)! = \frac{m!}{\binom{m}{k}} = (-1)^k m!$$

shows that (4.5) is independent of k and nonzero. Every unbalanced word has coefficient zero. This is the required multiple of (4.1). $\square$

Proof of Theorem 1.1. The radical in vertex degree $\mathbf{1}_E + \mathbf{21}_T$ is $u_T \mathcal{R}(E)$ by (3.3)–(3.4). Proposition 4.2 and Lemma 4.3 express each of its elements as a linear combination of

$$\phi(F_J) u_T z_M, \quad J \subseteq E, \quad |J| = 2m, \quad M \text{ a matching of } E \setminus J.$$

These are precisely products of the proposed generators with elements of C_N . Conversely, each $\phi(F_J)$ lies in the radical, which is an ideal. This proves (1.3), using $\ker \phi = Q_N$.

In ordinary degree k , a nonzero radical piece requires $s \geq m$ and $k = s + |T|$. There is therefore no radical in degrees below m . In degree m , the only radical pieces have $s = m$ and $T = \emptyset$. Each is a line, and the $\binom{N}{2m}$ different supports give distinct vertex degrees. The classes of the F_U form a basis in that degree and all are necessary as generators modulo Q_N . If $p \geq 5$,

then $m > 2$, so the quadratic part of the apolar ideal is exactly $(Q_N)_2$. If $p = 3$, the additional generators also have degree two. The assertions about minimal degrees and quadratic generation follow.

In characteristic zero Proposition 3.2 gives zero radical in every vertex degree. Lemma 3.1 then yields $\text{Ann}(D_N) = Q_N$. $\square$

5 Hilbert series and the first modular defects

Proof of Theorem 1.2. For ordinary degree k in (3.4), choose the $2s$ endpoints of E , then the $k - s$ vertices of T in its complement. Proposition 3.2 gives

$$\dim_K(A_N)_k = \sum_{s=0}^{\lfloor N/2 \rfloor} \binom{N}{2s} \binom{N-2s}{k-s} b_s(p). \quad (5.1)$$

Summing over k with weight t^k proves (1.4).

At $t = 1$ the sum is $\sum_s \binom{N}{2s} 2^{N-2s} (P^{2s})_{0,0}$. Odd powers of P have zero $(0,0)$ entry, so the binomial theorem gives $((2I + P)^N)_{0,0}$. This counts length- N walks in the strip $0, \dots, p-2$, allowing up and down steps and two colors of horizontal steps. For unrestricted walks, the number with displacement r is

$$[z^r](2 + z + z^{-1})^N = [z^{N+r}](1 + z)^{2N} = \binom{2N}{N+r}.$$

Reflection at the absorbing boundaries -1 and $p-1$ gives the image displacements $2pj$ and $2pj+2$, with opposite signs. Horizontal-step colors are preserved under reflection. Summing these contributions proves (1.5).

In characteristic zero, substitute $b_s(0) = \text{Cat}_s$ into (5.1). For $0 \leq k \leq N$, each summand becomes

$$\frac{N!}{s!(s+1)!(k-s)!(N-k-s)!}.$$

Factoring out $\binom{N}{k}/(k+1)$ leaves $\binom{k+1}{s+1} \binom{N-k}{s}$. Vandermonde's identity gives $\binom{N+1}{k}$ for the sum of these products. This is the first formula in (1.6). Summing over k gives $\binom{2N+2}{N}/(N+1) = \text{Cat}_{N+1}$. $\square$

Corollary 5.1. *The Hilbert function in characteristic $p > 2$ agrees with its characteristic-zero value for $N < 2p-2$. At $N = 2p-2$, its only change is a decrease by one in degree $p-1$.*

Proof. A Dyck path of semilength $s < p-1$ cannot reach height $p-1$. For $s = p-1$, exactly one path does: all up steps followed by all down steps. Therefore $b_s(p) = \text{Cat}_s$ for $s < p-1$ and $b_{p-1}(p) = \text{Cat}_{p-1} - 1$. The claim follows from (1.4). $\square$

Example 5.2 (Characteristic three). For $p = 3$, the allowed heights are 0 and 1, so $b_s(3) = 1$ for all s . Hence

$$H_{A_N}(t) = \sum_s \binom{N}{2s} t^s (1+t)^{N-2s}, \quad \dim_K A_N = \frac{3^N + 1}{2}. \quad (5.2)$$

At $N = 4$, the additional relation is

$$F_{\{1,2,3,4\}} = d_{13}d_{24} - d_{14}d_{23}.$$

It does not belong to Q_4 , since its exterior image is a nonzero multiple of Ω_2 . Direct differentiation over the integers gives

$$(d_{13}d_{24} - d_{14}d_{23})D_4 = 6(x_{13}x_{24} - x_{14}x_{23}). \quad (5.3)$$

This vanishes in characteristic three. The Hilbert function is $(1, 10, 19, 10, 1)$, compared with $(1, 10, 20, 10, 1)$ in characteristic zero. Thus the classical list of quadrics is incomplete in this characteristic, although the full ideal is still generated in degree two.

Example 5.3 (The first higher-degree generator). For $p = 5$ and $N = 8$, the first additional generator is the quartic

$$\det(d_{ij})_{1 \leq i \leq 4, 5 \leq j \leq 8}.$$

It is necessary modulo the quadratic ideal. The middle Hilbert value decreases from 1764 to 1763, and the total dimension decreases from $\text{Cat}_9 = 4862$ to 4861.

Corollary 5.4. *In every odd characteristic, the Hilbert function of A_N is symmetric and unimodal, with center $N/2$.*

Proof. Every polynomial $t^s(1+t)^{N-2s}$ in (1.4) has a symmetric unimodal coefficient sequence with center $N/2$, and its coefficient $\binom{N}{2s}b_s(p)$ is nonnegative. $\square$

6 Multiplication by a linear form

We first reduce the calculation to the diagonal sum ℓ and then describe its action explicitly. A symmetric matrix $H = (h_{ij})$ defines the directional differential operator

$$D_H = \sum_{i \leq j} h_{ij} d_{ij}.$$

Thus $D_H f(X)$ is the coefficient of t in $f(X + tH)$. This convention specifies the off-diagonal coefficients without a trace-pairing normalization.

Lemma 6.1. *Over an algebraic closure of K , all D_H with H nonsingular have the same ranks of multiplication by every power in A_N . These ranks are the generic ranks and are realized by $\ell = D_I$. If any linear form over any extension of K is weak or strong Lefschetz, then ℓ has the same property.*

Proof. Let $T_g f(X) = f(gXg^\top)$. Since $T_g D_N = \det(g)^2 D_N$, conjugation $f(\partial) \mapsto T_g^{-1} f(\partial) T_g$ preserves the apolar ideal. The chain rule gives $T_g^{-1} D_H T_g = D_{gHg^\top}$. Over an algebraically closed field of characteristic different from two, the nonsingular symmetric matrices form a single congruence orbit, containing I .

For each degree and each power, the maximal-rank locus is open, since the relevant matrix entries are polynomial functions of the coefficients of the linear form. The nonsingular orbit is open and dense, so its constant rank is the generic rank. There are only finitely many maps needed for either Lefschetz property. If their common maximal-rank locus is nonempty, it meets that orbit. Field extension preserves ranks, and ℓ is defined over K . $\square$

For a set J , let

$$B(J) = K[y_i : i \in J] / (y_i^2 : i \in J), \quad L_J = \sum_{i \in J} y_i.$$

When $|J| = M$, we also write B_M for this algebra. Grading shifts are written so that a vector in degree j of $B(J)(-s)$ has total degree $s + j$.

Proposition 6.2 (Boolean decomposition). *There is an isomorphism of graded $K[\ell]$ -modules*

$$A_N \simeq \bigoplus_{\substack{E \subseteq [N] \\ |E|=2s}} U(E) \otimes_K B([N] \setminus E)(-s), \quad (6.1)$$

where $U(E)$ is placed in degree zero before the shift, and ℓ acts on the summand indexed by E as $1 \otimes L_{[N] \setminus E}$.

Proof. Group (3.4) by its endpoint set E . The remaining vertex degrees are 0 or 2, indexed by subsets T of the complement of E . Multiplication by u_i adds i to T if it is not already present, and otherwise gives zero. It also gives zero for $i \in E$, because the vertex degree would become three. This is exactly the asserted action of the square-zero variables y_i . Each bracket matching on E has ordinary degree s , giving the shift. $\square$

The decomposition is a statement about a graded module and its linear operator. It is not an algebra decomposition. All of its summands have their Hilbert-function center at the same total degree $N/2$.

Define $r_M(j, q)$ to be the rank of multiplication by $(y_1 + \cdots + y_M)^q$ from $(B_M)_j$ to $(B_M)_{j+q}$, and set it equal to zero for degrees outside $0 \leq j \leq j+q \leq M$.

Proposition 6.3 (Ranks of all powers). *Suppose $\text{char } K = p > 2$. For valid degrees, $r_M(j, q) = 0$ if $q \geq p$. If $0 \leq q < p$, set $u = \min(j, M - j - q)$. Then*

$$r_M(j, q) = \sum_{\substack{0 \leq i \leq u \\ p \nmid \binom{u+q-i}{u-i}}} \left(\binom{M}{i} - \binom{M}{i-1} \right). \quad (6.2)$$

For every $k, q \geq 0$, the generic rank in A_N is

$$\text{rank}(\ell^q : (A_N)_k \rightarrow (A_N)_{k+q}) = \sum_{s=0}^{\lfloor N/2 \rfloor} \binom{N}{2s} b_s(p) r_{N-2s}(k-s, q). \quad (6.3)$$

Proof. In the squarefree monomial bases of B_M , multiplication by $(y_1 + \cdots + y_M)^q$ is $q!$ times the inclusion matrix from j -subsets to $(j+q)$ -subsets. If $q \geq p$, the power itself is zero, since $(y_1 + \cdots + y_M)^p = \sum_i y_i^p = 0$.

For $q < p$, the scalar $q!$ is a unit. Complementation of subsets transposes the inclusion matrix when needed and replaces j by $u = \min(j, M - j - q)$. Now $M \geq 2u + q$, so Wilson's rank formula [F, Theorem 4.1], with $a = u + q$ and $b = u$, applies and gives (6.2). Formula (6.3) follows from Proposition 6.2, Proposition 3.2, and the $\binom{N}{2s}$ choices of E . Lemma 6.1 identifies these with the generic ranks. $\square$

Proof of Theorem 1.3. Suppose first that $N \leq 2p - 2$. For any Boolean summand with $M \leq N$ and $j < M/2$, one has $j + 1 < p$. In the $q = 1$ case of (6.2), $u = j$ and the divisibility factors are $j + 1 - i$ for $0 \leq i \leq j$. All are nonzero modulo p , so the sum telescopes to $\binom{M}{j}$. Multiplication is injective on the increasing half of the Boolean algebra. The perfect pairing between complementary subsets gives surjectivity on the decreasing half. Because all shifted Boolean summands in (6.1) have center $N/2$, their direct sum has maximal rank in every degree. Thus ℓ is weak Lefschetz.

If $N \geq 2p - 1$, consider the summand with $E = \emptyset$, namely B_N . In it the element

$$\ell^{p-1} = (p-1)! \sum_{|T|=p-1} u_T$$

is nonzero, but is killed by ℓ . Its degree satisfies $p-1 < N/2$, and Corollary 5.4 gives $\dim(A_N)_{p-1} \leq \dim(A_N)_p$. Maximal rank at this degree would require injectivity, so ℓ fails the weak Lefschetz property. Lemma 6.1 then rules out every other linear form. At $N = 2p - 1$, formula (6.2) shows that the endpoint-free summand loses exactly the $i = 0$ contribution, of size one. Every other summand has $M \leq 2p - 3$ and its middle map has full rank. The asserted kernel dimension follows.

For strong Lefschetz, if $N < p$, every Boolean summand has $M < p$. For valid degrees in (6.2), the top indices of the binomial coefficients are less than p , so none is divisible by p ; also $q!$ is a unit. All powers have maximal rank. More explicitly, the rank is $\binom{M}{u}$, the smaller of the two dimensions. For each fixed pair of total degrees $k, k + q$, the side with smaller dimension is the same in every shifted Boolean summand, since their centers coincide. Summing therefore preserves maximal rank.

If $N \geq p$, each d_{ij}^p is zero in A_N : every variable in D_N has exponent at most two, strictly less than p . Frobenius gives $L^p = 0$ for every $L \in (A_N)_1$. But $(A_N)_N$ is nonzero, since $d_{11} \cdots d_{NN} D_N = 1$. Hence L^N cannot induce the required isomorphism $(A_N)_0 \rightarrow (A_N)_N$. Strong Lefschetz fails.

In characteristic zero, the subset-inclusion maps all have maximal rank. This follows, for example, from the same inclusion-matrix rank theorem: for each fixed matrix take a prime larger than M , for which the preceding binomial factors and factorials are units; full rank modulo that prime implies full rank over $\mathbb{Q}$. The module decomposition then proves that ℓ is strong Lefschetz. $\square$

Acknowledgment. The author acknowledges OpenAI GPT-6 for assistance with research exploration, proof development, verification code, and manuscript preparation.

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